

SYLLABUS :-

Physical Chemistry Lab 1 0-0-3 Prerequisite: Physical Chemistry 1 Measurement of (i) partial molar volume of the components at a given composition and (ii) excess molar volume in an aqueous solution of KCl; Phase diagram of an azeotropic mixture; Determination of the critical solution temperature of the phenol-water system; Liquid-vapor composition diagram by refractometry; Determination of the distribution coefficient of ammonia between chloroform and water; Determination of the formula of a complex by use of distribution law; Determination of rate constant of the hydrolysis reaction of ethyl acetate catalyzed by an acid (HCl); Determination of activation energy of the hydrolysis reaction of ethyl acetate catalyzed by an acid; Precipitation titration by conductometry; Determination of standard EMF of an electrochemical cell; Potentiometric titration of an acid by a base; Preparation of buffer solutions and determination of pKa of a weak acids; Determination of the isoelectric point (pI) of an amino acid by (a) pH-metric titration, (b) by colour matching of indicator; Books: 1. Experiments in Physical Chemistry, 2nd Edn, J. M. Wilson, R. J. Newcombe, A. R. Denaro, R. M. W. Rickett, Pergamon Press Ltd, 1968. 2. Experimental Physical Chemistry, V. D. Athawale, P. Mathur, New Age International Publishers, 2001. 3. Experimental Physical Chemistry: A Laboratory Textbook, A. M. Halpern, Prentice Hall, 2nd edition, 1997;